Course Code	AL352			
Course Title	Artificial Intelligence			
Credit Hours	3(2+1)			
Prerequisites by Course(s) and Topics	Data Structures and Algorithms Discrete Structures			
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Quizzes (15%), Assignments (15%), Class Participation (5%) Mid Term (20%), Final Term (45%)			
Course Instructor				
Course Coordinator	Dr. Mohsin Ashraf			
Office Hours				
Plagiarism Policy	All the parties involved in first cheating case will be awarded Zero for that evaluation.			
Current Catalog Description	This course will introduce the basics of artificial intelligence (AI), its scope and application domain. Topics such as knowledge representation, strategies for the State space search, Heuristic Search, first order logic, knowledge-based systems and state of the art computational techniques like neural networks, genetic algorithm will be covered during the course.			
Textbook (or **Laboratory Manual** for Laboratory Courses)	Artificial Intelligence: A modern Approach by Russell and Norvig			
Reference Material	Artificial Intelligence Illuminated by Ben Coppin			
CLO	**CLO STATEMENT**		**Bloom's Taxonomy Level**	**PLO**
1	To understand and explain the solid fundamental principles and concepts of artificial intelligence, including machine learning and neural networks.		C2	P1
2	To design and implement AI applications that solve real-world problems.		C3	P3
3	To critically analyze and differentiate the implications of AI technologies enabling them to make well-informed choices regarding the design and implementation of AI systems for classical search problems.		C4	P5
Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Attached			
Programming Assignments Done in the Course	Yes			
Class Time Spent on (in credit hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**

	1	0.5	0.5	
Oral and Written Communications				
Details of Technology Involved	Python, Pandas and NumPy, matplotlib, Pytorch, Anaconda			
Design Skills & Technologies Practiced	The AI undergraduate subject emphasizes blending technical proficiency with creative problem-solving to prepare students for AI design and development. Core skills include algorithm design, UI/UX, system architecture, and design thinking, with a focus on prototyping, testing, and scalability. Practical techniques cover model evaluation, version control, ethics, and effective communication, while project-based learning, and workshops provide hands-on experience. Students participate in FYP exhibitions for industry exposure and career opportunities. This holistic approach equips students to create innovative, user-friendly, and impactful AI solutions.			

Week	Lecture	Topics Covered		Evaluation Instrument	Relation to CLO
1	1	**Introduction to Artificial Intelligence**: History, Goals, Applications, Different AI Approaches, Turing Test, Cognitive models, Syllogism, Rational Agents **Intelligent Agents**, The Concept of Rationality, The Nature of Environments, The Structure of Agents	Chapter#1 Chapter#2		**CLO_1**
2	2	**Solving Problems by Searching**, Problem-Solving Agents, Problem Formulation, Search spaces, State space, Search Trees, Initial and Goal State, Successor Function, Objective Function, Goal Test, Goal Path. **Uninformed/Blind Search**: Breadth First Search (BFS), Depth First Search (DFS), Time Complexity, Space Complexity, Completeness, Optimality	Chapter#3		**CLO_1** **CLO_2**
3	3	**Uninformed/Blind Search**: Depth Limited Search, Uniform Cost Search, IDS, Algorithm Complexity	Chapter#3		**CLO_2** **CLO_3**
4	4	**Informed searches**: Heuristics, Heuristic Functions and Properties, Best First Search, Greedy Search **Informed Searches**: A* Search,	Chapter#3		**CLO_2**
5	5	**Informed Searches**: Monotonicity and Admissibility of Heuristics, Optimality vs Efficiency **Informed Searches**: IDS A*	 Chapter#3		 **CLO_2**
6	6	**Meta-Heuristics and Local Search**, Hill Climbing, Local minima and maxima, Variants of Hill Climbing Algorithm	Chapter#4		**CLO_2**

7	7	**Adversarial Search**: Game Trees, Zero-Sum Games, Minimax, Alpha beta pruning	Chapter#5		**CLO_2**
8	8	Review			
Midterm					
9	9	Genetic Algorithms, Representation, Fitness, Selection Mutation and Cross over, Solution of 8-queens, TSP through Genetic Algorithms, Knapsack	Chapter#4		**CLO_2** **CLO_3**
10	10	**Knowledge Representation (KR)**: Logic based, procedural, structural and Network representations, Frames **Propositional Logic**, Syntax, Connectives, Sentences, Ambiguity in Grammar, Semantics, Inference Rules	Chapter#7		**CLO_1** **CLO_2**
11	11	**FOL, Predicate Logic**, Problems with propositional logic, Predicates, Universal Quantifiers, Existential Quantifiers	Chapter#8		**CLO_2**
12	12	**Introduction to Artificial Neural Networks**, Supervised and Unsupervised Learning, Perceptron AND, OR etc. classification through Perceptron Performance measures: Confusion Matrix (Binary & Multi Class) − Evaluation metric: Accuracy, recall. Precision, F1 score	Chapter#18		**CLO_1** **CLO_2**
13	13	**Supervised Learning: Bayesian** Classification, Naïve Bayes Theorem and examples, Conditional Independence of attributes **Decision Trees** Convolutional Neural Network (fwd prop only) Activation Functions (Linear,Sigmoid, Relu, tanh, Softamax)	Chapter#18		**CLO_2** **CLO_3**
14	14	K-NN, for classification example	Chapter#18		**CLO_2** **CLO_3**
15	15	Clustering with K-means	Chapter#18		**CLO_2** **CLO_3**
16	16	Review			